Zengxiang Lei

Transportation AI, Simulation, and Cyber-Physical Systems Researcher

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Professional Summary

I build scalable models and decision systems for ride-hailing, shared autonomous electric fleets, evacuation, and transportation security.

Experience

Transportation Cybersecurity and Resiliency Center, Purdue University, Postdoctoral Associate — West Lafayette, IN — Jan 2025 – present — 1 year 10 months

- Lead research connecting METS-R SIM, CARLA, and an autonomous driving stack for controlled, repeatable scenario testing.
- Develop amortized samplers for policy and control verification in simulated environments, with emphasis on rare, high-impact scenarios.
- Study adversarial risks in autonomous mobility systems, including booking attacks, emergent adversarial driving maneuvers, and cybersecurity challenges for next-generation road transportation.

Urban Mobility, Network and Intelligence Lab, Purdue University, Graduate Research Assistant — West Lafayette, IN — Aug 2018 – Dec 2024 — 6 years 5 months

- Conducted research on pricing and fleet relocation, leading to publications in Transportation Research Part B and Transportation Research Part C.
- Served as student lead for the U.S. Department of Energy METS-R project and contributed substantially to METS-R SIM, a city-scale multimodal electric vehicle simulator.
- Analyzed large-scale GPS and trajectory data to study evacuation, return, pandemic-era mobility, and transportation hub operations.

Education

Ph.D. **Purdue University**, Transportation and Infrastructure Systems Engineering — West Lafayette, IN — 2018 – 2024

- Dissertation: Towards an Efficient and Secure Ride-hailing Service.
- Selected coursework: Reinforcement Learning, Real Analysis and Measure Theory, Security Analytics.

M.S.E. and B.S. **Tongji University**, Transportation Planning and Management; Transportation Engineering — Shanghai, China — 2011 – 2018

Selected Research and Open-Source Leadership

METS-R SIM: Open-Source Multimodal Electric Mobility Simulator

U.S. Department of Energy project

- Led development of a city-scale simulation platform for Multi-modal Energy-optimal Trip Scheduling in Real-time, supporting shared autonomous electric vehicles, taxis, buses, charging operations, routing, and service evaluation.
- Used the platform to evaluate energy, congestion, and service impacts of operational algorithms for electric mobility services and transportation hubs.

Secure and Efficient Ride-Hailing Systems
Ph.D. research

- Designed scalable reinforcement learning approaches for dynamic pricing and data-driven relocation policies for mobility-on-demand systems.
- Quantified the network-level risk of adversarial booking attacks against autonomous mobility-on-demand services.

Large-Scale Mobility Analytics for Disasters and Recovery
Interdisciplinary research

- Modeled evacuation and return behavior during Hurricane Ida under compound risks with COVID-19.
- Built evidence from smartphone GPS and trajectory data to study transportation resilience, demand recovery, and traveler response.

Honors and Awards

Eldon J. Yoder Memorial Award 2024
Purdue University, 2024-2025 academic year.

NSF Innovation Corps Midwest Industry Connect Grant 2019
National Science Foundation I-Corps Midwest. Awarded for work on social ride-sharing.

Chinese Patent: A Method for Evaluating Reserve Capacity of Urban Metro Networks 2018

Peikun Transportation Talent Fund Award 2017
Tongji University.

13th Competition of Transport Science, First Prize 2014
Project: Research on Spatial Activity Patterns Based on Smartphone Data.

Selected Publications

- Lei, Z., Shreekumar, A., Rosenthal, J., Song, R., Cardenas, A. A., Fremont, D. J., Xu, D., Ukkusuri, S. V., and Celik, Z. B. (2026). Stable GFlowNets with TV Monitoring and Probabilistic Guarantees. Advances in Neural Information Processing Systems (NeurIPS), accepted.
- Cai, Z. J., Zhang, Y. F., Tan, R., Lei, Z., and Ukkusuri, S. V. (2026). Controllable Road Marking Generation. Advances in Neural Information Processing Systems (NeurIPS), accepted.
- Tan, R., Lei, Z., and Ukkusuri, S. (2026). Worst-Case Hidden-Vehicle Trajectory Search in Spatiotemporal Occlusion Regions. ECCV 2026 Workshop on Safe and Defensive Autonomous Driving (SDAD), accepted oral presentation and Best Paper Award.
- Yang, Z., Lei, Z., and Ukkusuri, S. (2026). RiskLens: Auditing the Causal Geometry of End-to-End Driving Policies. 1st IJCAI Workshop on Safe Physical AI.
- See the complete publication list for journal articles, conference papers, and preprints.

Invited Talks and Seminars

- "Transportation Cybersecurity: A Network-Level Perspective." TraCR Scholar Webinar Series, August 27, 2025.
- "Adversarial Booking Attacks for On-Demand Mobility Services." Purdue Institute of Transportation Engineers Student Chapter Seminar, November 28, 2023.

Teaching and Mentorship

Co-Instructor, Fall 2026 CCE564: Data Science for Smart Cities: Delivering half of the lectures and developing instructional materials on generative models for smart cities at Purdue University.

Co-Instructor, Fall 2025 CCE564: Data Science for Smart Cities: Delivered four weeks of lectures and developed instructional materials on deep learning and reinforcement learning for transportation systems at Purdue University.

Research Mentorship: Mentored seven students and junior collaborators: Mithun Debnath (2022), Ruichen Tan (2023–Present), Hongli Li (2024–Present), and Zhiyu Joey Cai, Ziyuan Yang, Shiveswarran Ratneswaran, and Yufan Flynn Zhang (2025–Present). Projects span generative AI, autonomous-system evaluation, mobility analytics, and transportation security.

Selected Achievements of Mentored Students

- Ruichen Tan (first author): Worst-Case Hidden-Vehicle Trajectory Search in Spatiotemporal Occlusion Regions — Best Paper Award and accepted oral presentation, ECCV 2026 Workshop on Safe and Defensive Autonomous Driving (SDAD).
- Zhiyu Joey Cai (first author): Controllable Road Marking Generation — accepted to the NeurIPS 2026 main track.
- Hongli Li (first author): Anticipatory demand planning for microtransit with strategic use of latent capacity at fixed stops — Transportation Research Part E: Logistics and Transportation Review, 212 (2026), 104939.
- Ziyuan Yang (first author): RiskLens: Auditing the Causal Geometry of End-to-End Driving Policies — 1st IJCAI Workshop on Safe Physical AI, 2026.
- Mithun Debnath (first author): Ride-Hailing Expansion and Urban Traffic Safety: A Data-Driven Perspective — accepted by ASCE's Journal of Transportation Engineering.

Professional Service

Journal Reviewer: IEEE Transactions on Information Forensics and Security; IEEE Transactions on Intelligent Transportation Systems; Transportation Research Parts B, C, D, and E; Transportation; Multimodal Transportation; Transportation Research Record; Data Science for Transportation; European Journal of Operational Research; Operations Research Perspectives; Scientific Reports; Energy.

Conference Reviewer: AAAI Conference on Artificial Intelligence; Conference on Neural Information Processing Systems; ACM SIGKDD Conference on Knowledge Discovery and Data Mining; IEEE Intelligent Vehicles Symposium; IEEE Intelligent Transportation Systems Conference; IEEE International Conference on Vehicular Electronics and Safety.

Event Coordinator: ITE Special Seminar (2025), "Interdisciplinary Intelligent Transportation Systems: Traffic Control and Travel Time Dynamics," speaker Dr. Pushkin Kachroo.

Skills

Research Areas: Transportation AI, simulation-based verification, mobility-on-demand, shared autonomous electric vehicles, transportation cybersecurity, disaster mobility.

Methods: Reinforcement learning, agent-based simulation, large-scale mobility analytics, scenario search, trajectory data analysis.